# Muhammad Zawad Mahmud

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LinkedIn · Google Scholar · GitHub · Website


## RESEARCH INTERESTS

Computer Vision · Visual Place Recognition · Novel View Synthesis · Agentic AI · Machine Learning Security

## EDUCATION

**M.S. in Computer Science**, Fordham University, USA 2025 – May 2027 (expected)
CGPA 4.00 / 4.00 · Enrolled in MS Thesis

**B.S. in Computer Science & Engineering**, North South University, Bangladesh 2019 – 2023
CGPA 3.71 / 4.00 · Top 5% of class · Magna Cum Laude · Graduated with Academic Distinction

## SELECTED PUBLICATIONS

**1.** *Hallucination-Aware Multi-View Fusion with Foundation Models: Robust Learning from Low-Quality Synthetic Views.* **M. Z. Mahmud**, D. Lyons, M. Rahouti, D. Camacho, A. Chehri. **Under review, Information Fusion**

**2.** *Evaluating Place-Discriminative Representations in Synthetic Novel Views.* **M. Z. Mahmud**, S. Islam, D. Lyons, M. Rahouti, A. Chehri, K. Xiong. **Preprint, Arxiv**

**3.** *Survivor Takes All: Keep-Rule Substitution Attacks on Near-Duplicate Deduplication in LLM Pretraining.* **M. Z. Mahmud**, S. R. Alve, L. Khan. **Accepted, IEEE BigData 2026**

**4.** *Membership Privacy Auditing for Machine Learning: Selection-Aware Multi-Attack Evaluation and DP-SGD Analysis.* **M. Z. Mahmud**, S. Islam, M. Rahouti, D. Verma, P. Novotny, A. Chehri. **Under review, SN Computer Science**

**5.** *Interpretable Dengue Detection Using XAI-Integrated Ensemble and Large Language Models on a Reduced-Feature Clinical Dataset.* **M. Z. Mahmud**, M. S. Reza, S. Islam, S. R. Alve, R. Khan. **Array**, vol. 31, art. 101120, 2026.

**6.** *Advanced Vision Transformers and Open-Set Learning for Robust Mosquito Classification.* A. A. J. Karim, **M. Z. Mahmud**, R. Khan. **PLoS Computational Biology**, 20(12), e1012654, 2024.

**7.** *FairRate: Examining Responsibility and Rating Correction in Ride-Hailing Reputation Systems.* J. Bieniek, M. I. Mahmud, M. Z. Maqsood, **M. Z. Mahmud**, S. Islam, D. C. Verma, P. Novotny, M. Rahouti. **Under review, ACM FAccT 2027**

## RESEARCH EXPERIENCE

**Graduate Research Assistant**, Dept. of Computer & Information Science, Fordham University Aug 2025 – Present
*Advisors: Dr. Mohamed Rahouti, Dr. Damian Lyons*

- Lead first-author work on generative view augmentation for visual place recognition: a training-free method that ranks synthesized views by DINOv2 semantic drift and admits them under a fixed budget, beating random selection in all 24 wide-viewpoint comparisons across three datasets and four descriptors.
- Developed a synthetic novel-view augmentation pipeline for VPR under viewpoint/appearance shift.
- Disentangling representation robustness from reference-map coverage in visual place recognition through counterfactual-gallery evaluation.

**Research Assistant**, North South University, Bangladesh Jan 2023 – Aug 2025

*Advisors: Dr. Riasat Khan, Dr. Md. Mamun Molla*

- Produced applied-ML and explainable-AI work on clinical/biomedical classification, resulting in multiple peer-reviewed publications.
- Mentored undergraduate researchers on Disease Classification.

## TEACHING EXPERIENCE

- **Graduate Teaching Assistant**, Dept. of Computer & Information Science, Fordham University (Aug 2026 – Present).
- **Lab Instructor**, Dept. of Electrical & Computer Engineering, North South University (Feb – Aug 2025).
- **Adjunct Lecturer**, Dept. of Computer Science & Engineering, Green University of Bangladesh (Feb 2024 – Jun 2025); Structured Programming, Data Structures, DBMS, Data Communication.
- **Undergraduate Teaching Assistant**, North South University (Jan 2022 – Jan 2024).

## ADDITIONAL PUBLICATIONS

**First-author**

*Advance Transfer Learning Approach for Identification of Multiclass Skin Disease with LIME Explainable AI.* **M. Z. Mahmud**, M. S. Reza, S. R. Alve, S. Islam, N. Fahmid. IEEE International Conference on Computer and Information Technology (ICCIT) 2024

*Optimized IoT Intrusion Detection using Machine Learning.* **M. Z. Mahmud**, S. Islam, S. R. Alve, A. J. Pial. IEEE International Conference on Robotics, Automation, Artificial-Intelligence and Internet-of-Things (RAAICON) 2024

*Predicting Adolescents' Attitudes Toward COVID-19 Vaccination Using Machine Learning.* **M. Z. Mahmud**, I. Jahan, S. R. Alve, M. S. Reza, S. Islam. IEEE International Conference on Quantum Photonics, Artificial Intelligence & Networking (QPAIN) 2025

**Co-author**

*Enhancing Multi-Class Disease Classification Using Advanced LLMs.* A. A. J. Karim, **M. Z. Mahmud**, S. Islam, A. Azam. IEEE International Conference on Computer and Information Technology (ICCIT) 2024

*Deep Learning Approach for Oral Squamous Cell Carcinoma with LIME Explainable AI.* S. Islam, **M. Z. Mahmud**, S. R. Alve, M. M. U. Chowdhury, F. I. Oishe. IEEE International Conference on Computer and Information Technology (ICCIT) 2024

*Deep Learning Driven Respiratory Sound Analysis for Early Chronic Obstructive Pulmonary Disease Detection,* S. R. Alve, **M. Z. Mahmud**, M. A. Islam, S. Islam, S. Rahman, M. M. Khan. IEEE International Conference on Quantum Photonics, Artificial Intelligence & Networking (QPAIN) 2026

*Resource-Efficient Machine Learning for Multi-Class Threat Detection in IoT Environments.* S. R. Alve, **M. Z. Mahmud**, S. Islam, Md. A. Chowdhury, J. Islam. IEEE International Conference on Quantum Photonics, Artificial Intelligence & Networking (QPAIN) 2025

*Artificial Intelligence Driven Intrusion Detection for SDN Environments.* S. Islam, S. R. Alve, **M. Z. Mahmud**, F. I. Oishe, M. M. Khan. Electrical, Computer, and Communication Engineering (ECCE) 2025

## TECHNICAL SKILLS

**Languages:** Python, C++, C
**ML / DL:** PyTorch, TensorFlow, Keras
**Computer Vision / 3D:** OpenCV, COLMAP, NeRF & Gaussian-Splatting frameworks, Open3D.
**Tools / Infra:** Linux, Git, Docker, HPC/SLURM.

## AWARDS & STANDARDIZED TESTS

- Merit-Based Graduate Scholarship, Fordham University.
- 75% Merit Financial Aid, undergraduate (North South University).
- Magna Cum Laude (undergraduate CGPA ≥ 3.65).
- GRE 321 (V 159, Q 162, AWA 4.5)
- DET 130 (L 130, R 115, S 135, W 135)
- TOEFL 107 (L 30, R 26, S 24, W 27) [Expired]

## ACADEMIC SERVICE

**Reviewer:** IEEE Transactions on Network and Service Management; Internet of Things & Cyber-Physical Systems (Elsevier); Next Material Journal (Elsevier); Array (Elsevier); Meteorological Applications (Wiley); IEEE Conference on Dependable and Secure Computing (DSC 2026); Intl. Conf. on Intelligent Data Analysis and Applications (IDAA 2025); International Conference on Electrical and Computer Engineering Researches (ICECER 2026); International Conference on Computing Advancements (ICCA 2026).